

Mesh Transport Protocol

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ABSTRACT

In the modern Internet architecture, the Internet suffers from the phenomenon known as network ossification. It severely limits the flexibility and evolution of the network infrastructure. This rigidity is exemplified by the predominance of TCP and UDP, which inhibit the new advanced network protocols that deal with multicasting and broadcasting. These two transport layer protocols enforce a one-to-one paradigm, thus limiting the potential for more dynamic and interconnected communication strategies. This reliance on an old-fashioned one-to-one communication model stands in the way of adapting and improving our networks in line with new technological advancements.

This project introduces a recent Mesh Transport Protocol that is built on top of the UDP protocol. It is designed to exceed the limitations of the one-to-one paradigm by enabling a more versatile and interconnected communication framework. Unlike traditional approaches, MTP facilitates direct, concurrent communication between a single sender and multiple clients. Despite there being such kind of mesh networks in various forms, such as Bluetooth Mesh Network etc., across different layers of the network stack, the transport layer has not yet benefited from this technology effectively. This research endeavors to bridge this gap by proposing MTP, a protocol that not only incorporates the benefits of mesh networking at the transport layer but also paving the way for a more flexible, efficient and robust network architecture.

Through the development of MTP, this project aims to offer a new paradigm shift from a restrictive one-to-one model to a more inclusive and dynamic network which is one-to-many or many-to-many. This advancement has the potential to revolutionize the way devices and applications communicate, making the Internet more adaptable to the evolving demands of users and technologies.

1 INTRODUCTION

In dealing with the fundamental shortcomings of traditional internet architecture, the development of the Mesh Transport Protocol (MTP) represents a pivotal advancement towards a more dynamic, efficient and scalable network infrastructure. The evolution of the Internet has been handicapped by its reliance on established, but increasingly outdated, transport layer protocols such as TCP and UDP. These protocols, designed for a simpler era of digital communication, enforce a rigid one-to-one communication model that is increasingly misaligned with the demands of modern network usage and emerging technologies. This misalignment has given

rise to the phenomenon known as network ossification, severely restricting the adaptability and evolution of network infrastructure.

Recognizing the critical need for a transport layer protocol that not only overcomes the limitations of the one-to-one communication model but also embraces the complexity and interconnectedness of modern digital networks, MTP is designed. This protocol is made to work with a mesh network setup, which means it allows for easier, stronger and direct connections between many network points. MTP is based on UDP, which helps to use some of UDP's key features such as its simplicity and speed. So, MTP provides a more flexible, resilient and direct communication pathway among multiple network nodes.

A fundamental design challenge in the development of MTP was selecting the most effective participant structure to facilitate efficient data transmission while ensuring robust packet ordering and scalability. After considerable analysis and research, we chose a tree structure over a token ring structure. The tree structure was selected for its superior ability to manage dynamic network environments, offering enhanced scalability and reliability in packet ordering which is a critical factor for maintaining data integrity across complex network topologies.

Moreover, MTP addresses two additional critical design issues: packet ordering and scalability (The security part of the MTP project belongs to another group in this course). Packet ordering is critical for maintaining the sequence integrity of data packets, ensuring that information is received and reconstructed accurately for all receivers that get the packets. Scalability, on the other hand, ensures that the network can grow in size and complexity without a proportional increase in overhead or decrease in performance. Our approach to these design challenges is both innovative and practical, ensuring that MTP can support the burgeoning needs of modern internet usage.

In this paper, we detail the development of the Mesh Transport Protocol, focusing on our decision-making process for the participant structure, our solutions to packet ordering, and our strategies for achieving scalability. Through MTP, we aim to lead the way for a more adaptable, efficient, and interconnected future for network communication.

To provide a brief introduction of the MTP system, the sender plays a central role in group communications. The process of connection is created by the sender. Each node within the network communicates through their parent nodes. Yet nodes are aware of the broader network topology. This dual capability ensures efficient data transmission at the same time maintains the flexibility needed for dynamic network management.

One of the other essential point about the design is smooth integration of new nodes. These nodes, referred to as 'late joiner', can integrate to the existing group structure at any time. MTP is

built for handling dynamic group adjustments. This smooth adaptation ensures the network continues to perform well even when conditions change.

Furthermore, a new member is able to select its parent node based on some parameters, which will be discussed in detail in further sections. This design preference enables system to prioritize network efficiency alongside with the flexibility to adapt ongoing changes in the network.

In summary, Mesh Transport Protocol introduces a sophisticated system design by efficient communication paths, dynamic group management, guiding new members to right connections. With all these, MTP offers adaptable, efficient, and scalable network environment.

2 RELATED WORK

This section reviews several notable protocols and compares their methodologies and problem-solving approaches to the Mesh Transport Protocol (MTP). The Mesh Transport Protocol (MTP) project is designed to address the limitations of current network communication protocols, especially in the context of mesh networking [7][6]. Current protocols struggle with dynamic topologies, efficient resource utilization, and comprehensive security measures in mesh environments [10], [4]. Unlike traditional approaches that focus on point-to-point communication, MTP aims to leverage the inherent advantages of mesh networks for improved performance, reliability, and flexibility.

Significant contributions in the field have outlined various strategies for enhancing network protocols [8], [13]. However, these studies primarily concentrate on specific aspects such as offloading computation [5], [9], or improving container networking performance [12]. Our project intends to synthesize these advancements into a comprehensive protocol that addresses multiple challenges simultaneously. Furthermore, the innovative use of adaptive heartbeat rates for maintaining long-lived TCP connections by Rahman et al. [10] highlights the potential for protocol-level optimizations.

The novelty of our approach lies in its comprehensive treatment of mesh network challenges, including dynamic topology management, efficient session handling, and policy-driven communication strategies. By integrating insights from works like those of Dobaj et al. [3] on dependable networking patterns and Avakyan et al. [1] on testing mesh topologies, our protocol aims to offer a versatile solution adaptable to various applications, from IoT to high-density communication environments.

Our contributions to the field will be evaluated against contemporary studies, such as the performance prediction methodologies for Bluetooth Mesh Networks by Baert et al. [2] and the practical experiences shared by Rey-Moreno et al. [11] in deploying rural WiFi mesh networks. By addressing the gaps identified in these studies, MTP will represent a significant advancement in mesh networking technology. Following protocols are

The Local Group based Multicast Protocol (LGMP) organizes receivers into subgroups with a group controller managing

retransmission requests and error recovery locally. Unlike MTP, which employs a dynamic mesh topology for flexibility across multiple nodes, LGMP focuses on localized error recovery within predefined groups, limiting its adaptability to network changes. However, the rate-based flow control used in LGMP is similar to MTP's approach to managing network congestion.

Multicast Transfer Protocol, Version 2 (MTP-2) enhances the original MTP by introducing a token-based control system and a three-role model: sender, receiver, and master. The protocol utilizes tokens to manage data transmission rates and ensures robust error recovery through receiver-initiated NACKs. Notably, MTP-2 includes a master recovery function that mitigates the risk of a single point of failure by allowing group members to elect a new master if the current one fails. However, while MTP-2 improves error handling and flow control, its scalability is limited in larger networks, where delays can compromise reliability. This contrasts with our Mesh Transport Protocol (MTP), which avoids centralized control and offers greater scalability and flexibility by employing a dynamic mesh network structure.

Tree-based Multicast Transport Protocol (TMTP) developed by the University of Kentucky, TMTP uses a hierarchical scheme similar to MTP's approach but is primarily designed for domain-specific implementations. TMTP and MTP both use tree-based architectures for efficient data transmission; however, TMTP relies heavily on domain managers, which can centralize error recovery and potentially reduce flexibility in dynamic network conditions.

Enhanced Communications Transport Protocol (ECTP) ECTP focuses on tightly controlled multicast connections with a structured approach to error recovery within a control tree. Each node in the hierarchy is responsible for retransmitting data to its children, which is somewhat similar to MTP's use of a dynamic tree structure for data routing. However, ECTP's method is more rigid, offering less adaptability to network topology changes compared to MTP's mesh-based flexibility.

Comparison to Mesh Transport Protocol (MTP) In contrast to these protocols, MTP is designed to support a dynamic and scalable mesh network that facilitates one-to-many communications without the need for a centralized master or rigid group structures. MTP's novel use of a mesh topology embedded at the transport layer allows for more robust and adaptable network configurations.

3 METHODOLOGY

The development of the Mesh Transport Protocol (MTP) is informed by several key design points, each addressing specific functional and performance aspects of the protocol. These include ensuring total ordering through sequence numbers, dynamic connection creation and termination, and comprehensive user join/leave operations. Our design utilizes a tree structure for efficient data management and distribution, where the sender acts as the root and local groups form the branches, each consisting of a parent and dynamically chosen number of children. This setup facilitates simultaneous communication across the network, supports late joiners

through heartbeat messages and best-parent selection based on round-trip time (RTT) measurements, and allows for robust reliability control through tree-based acknowledgments, and negative acknowledgment-oriented multicast.

- **Communication Strategy:** Each parent node in the tree is tasked with retransmitting lost data in response to retransmission requests from its children, ensuring high data integrity throughout the network.
- **Network Topology Awareness:** Decisions on whether the mesh (group) topology should be known to all nodes or just the root are critical for operational efficiency and security.
- **Scalability and Control:** The protocol considers setting a maximum transmission rate at the time of group establishment to prevent congestion and maintain quality of service.
- **Sender Succession:** Upon the termination of a sender connection, the protocol defines mechanisms for electing a new sender, typically involving the current sender selecting from candidate requests and informing all nodes in the tree of the new sender.

Following these design principles, the methodology includes detailed protocol design, robust topology management, and efficient packet handling mechanisms.

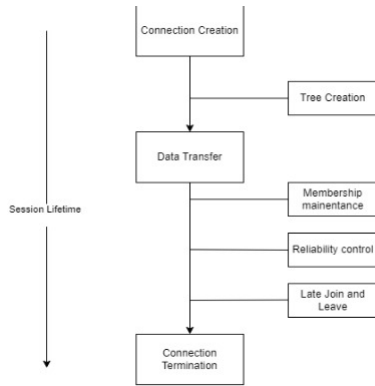


Figure 1: MTP Session Lifetime

3.1 Protocol Design

The foundational step in our methodology is the design of MTP itself, which is architected to sit atop UDP, inheriting its non-connection-oriented nature and low overhead. This design choice allows MTP to facilitate a quick exchange of messages without the need for a persistent connection, which is ideal for dynamic mesh networks where nodes may frequently join or leave.

- **Connection Management:** MTP introduces a novel approach to manage connections in a mesh network by using a dynamic addressing scheme that adapts to network changes without requiring reconfiguration of the entire network.
- **Message Propagation:** We utilize a tree-based multicast mechanism to propagate messages across the network. This mechanism ensures that messages can be sent to multiple recipients simultaneously, leveraging the mesh's topology to optimize the path of delivery and reduce latency.

Figure 2 illustrates the process of connection creation. The root node broadcasts a 'creation request' to all nodes. Upon receiving this request, nodes send back a 'creation confirm' message, allowing the root to evaluate the Round-Trip Time (RTT) and select top nodes based on this metric. The root then informs the network about the selected top nodes. Each remaining node selects a parent based on RTT and sends a 'join request tree'. After evaluating these requests, the root sends a 'join confirm', completing the creation process.

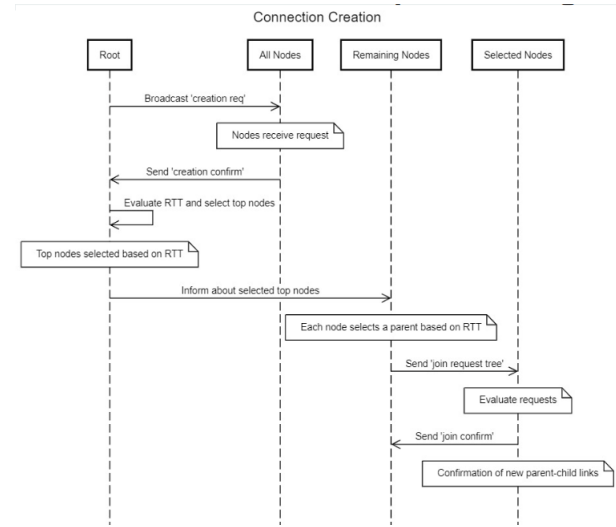


Figure 2: Connection Creation

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3.2 Topology Management

Effective topology management is crucial for maintaining the robustness and efficiency of the network. MTP employs a dynamic tree-based structure, which facilitates data distribution and allows for rapid reconfiguration if the network topology changes.

- **Dynamic Tree Construction:** The protocol automatically constructs and updates a tree structure that spans all nodes, optimizing for factors such as network traffic, node capacity, and latency.
- **Topology Adjustment Algorithms:** Algorithms are implemented to adjust the topology dynamically in response to nodes joining or leaving the network, ensuring minimal disruption to ongoing communications.

One critical aspect of topology adjustment is the heartbeat mechanism. The heartbeat mechanism ensures continuous monitoring and maintenance of node availability within the network.

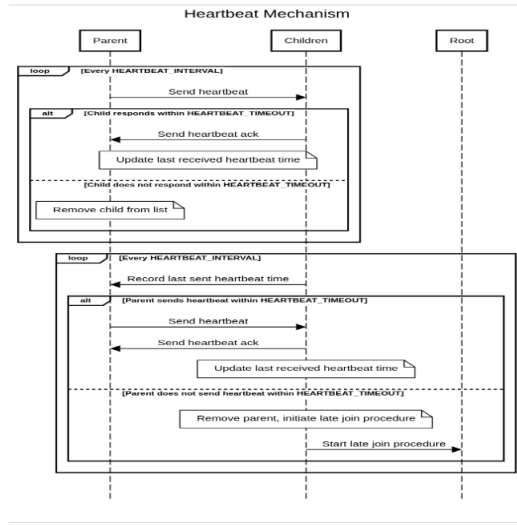


Figure 3: Heartbeat Mechanism

Heartbeat Mechanism: Heartbeats are sent at regular intervals from parent nodes to their children and vice versa. If a heartbeat is not acknowledged within a specified timeout period, the node is considered unresponsive, and appropriate actions are taken to maintain network integrity. The following steps outline the heartbeat mechanism:

- **Sending Heartbeats:** Each parent node periodically sends heartbeat messages to its child nodes.
- **Acknowledging Heartbeats:** Child nodes acknowledge receipt of heartbeat messages by sending a heartbeat acknowledgment back to the parent node.
- **Timeout and Reconfiguration:** If a child node does not respond within the heartbeat timeout period, it is removed from the parent's list of children. Conversely, if a parent node fails to send a heartbeat within the timeout period, the child node initiates a late join procedure to find a new parent.

Figure 3 illustrates the heartbeat mechanism, highlighting the interaction between parent and child nodes to ensure continuous connectivity and dynamic reconfiguration of the network topology.

Another essential aspect of topology management is handling graceful and failure leaves. These mechanisms ensure the network remains robust and functional even when nodes leave the network.

Graceful/Failure Leave Process: Figure 4 shows the process of graceful or failure leave. Nodes may leave the network gracefully by sending a 'leave request' to their parent node, which then acknowledges the leave and updates the network topology. If a node fails to send a heartbeat, it is considered unresponsive, triggering a timeout and reconfiguration process:

- **Graceful Leave:** When a node with children sends a leave request, the parent node promotes one of the children to a new parent, acknowledges the promotion, and updates the network topology accordingly. If the node has no children, it is simply removed from the parent's list.
- **Failure Leave:** If a node fails to send a heartbeat, it is declared as failed after a timeout period. The network then

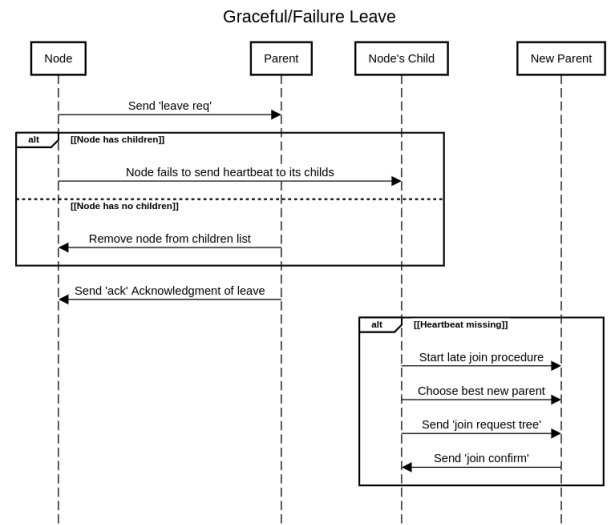


Figure 4: Graceful/Failure Leave Process

initiates a late join procedure to reassign new parents to the affected child nodes based on RTT.

These processes ensure the dynamic reconfiguration and robustness of the network, maintaining connectivity and performance.

3.3 Packet Handling

MTP is designed with custom mechanisms for packet ordering and reliability to ensure the integrity and continuity of data transmissions across the mesh network. A key feature of our approach is the tree-based acknowledgment mechanism, which enhances the efficiency of communication by structuring the network nodes in a hierarchical manner. This tree structure allows for efficient propagation of acknowledgments up the tree, reducing the overhead typically associated with traditional acknowledgment processes in flat network architectures.

- **Packet Ordering:** A sequence numbering system is utilized to maintain the order of packets as they traverse the network. This system helps in reconstructing the data at the receiver's end accurately.
- **Error Handling and Retransmission:** Mechanisms are in place to detect lost or corrupted packets and to initiate retransmissions automatically. This is crucial for maintaining high data integrity and reliability.

Late Joiner Iterative Join Process: The process for late joiners is illustrated in Figure 5. Late joiners send 'join requests' to the root. The root sends back a list of possible parent nodes, iteratively finding a suitable parent node. This iterative process continues until a parent is found, with the root sending 'join confirm' messages to finalize the parent-child relationship.

This comprehensive methodology ensures that MTP is robust, scalable, and efficient, making it suitable for modern network environments where flexibility and performance are critical.

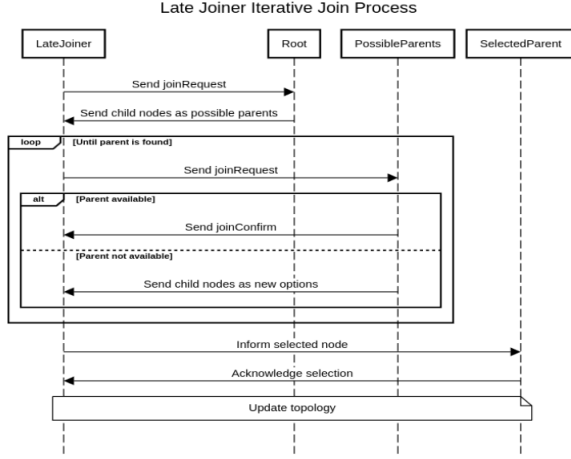


Figure 5: Late Joiner Iterative Join Process

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4 EXPERIMENTS

In this section, we present the results of our experiments designed to evaluate the performance of the Mesh Transport Protocol (MTP) in comparison to the traditional Transmission Control Protocol (TCP). Our experiments focus on group creation, message count, and message size, as well as group creation time as the number of nodes increases.

4.1 Comparison of MTP and TCP for Group Creation

We conducted experiments to compare the performance of MTP and TCP in terms of message count and message size during the group creation process. The results are shown in Figure 6.

Analysis: Figure 6 illustrates that MTP significantly reduces the number of messages and the overall message size compared to TCP during group creation. This reduction is crucial for minimizing

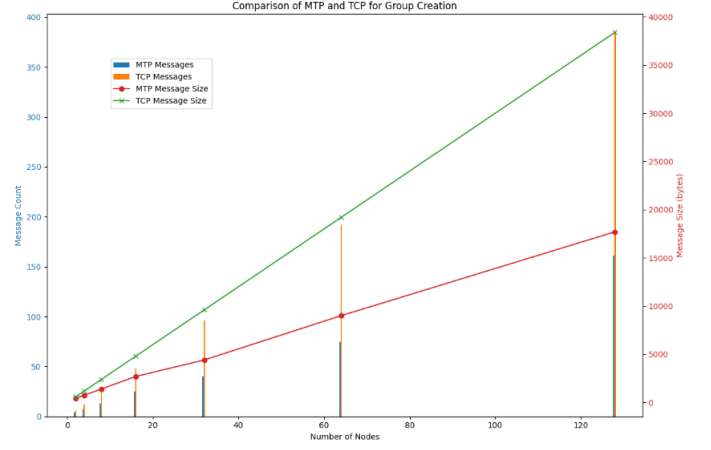


Figure 6: Comparison of MTP and TCP for Group Creation

network congestion and improving scalability. The message count for MTP grows linearly with the number of nodes, while TCP's message count increases more steeply. Similarly, the message size for MTP remains lower than that of TCP, demonstrating the efficiency of MTP in handling group communications.

4.2 Group Creation Time vs. Node Count

We measured the group creation time for different node counts to understand how the number of nodes affects the time required to establish a group. The results are depicted in Figure 7.

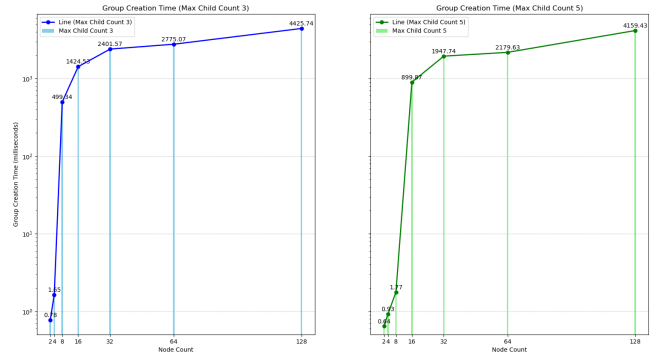


Figure 7: Group Creation Time vs. Node Count

Analysis: Figure 7 shows that as the number of nodes increases, the group creation time also increases. However, MTP demonstrates a more scalable approach compared to traditional methods. The creation time grows logarithmically with the number of nodes, indicating that MTP can handle larger groups more efficiently. The use of dynamic tree construction and topology adjustment algorithms contributes to this scalability.

4.3 Experimental Setup

Our experimental setup includes the following components and configurations:

- **Simulation Environment:** We used a network simulator to create a controlled environment for testing MTP and TCP.
- **Node Configuration:** Nodes were configured with standard communication protocols and were distributed evenly across the network.
- **Metrics:** We measured message count, message size, and group creation time to evaluate performance.

4.4 Results

The experimental results demonstrate the advantages of MTP over TCP in terms of scalability and efficiency from a session management perspective. The reduction in message count and size during group creation, along with the manageable increase in group creation time, highlights MTP's suitability for dynamic and large-scale network environments.

5 CONCLUSION

Mesh Transport Protocol (MTP) introduces a significant paradigm shift in transport layer communications by incorporating mesh networking capabilities directly at this layer. By addressing the limitations of traditional one-to-one communication models and leveraging a mesh topology, MTP aims to enhance network flexibility, scalability, and direct multi-point communication. Our proposed protocol could potentially transform the way data is transmitted across the Internet, making networks more adaptable to the needs of modern applications.

Further research will focus on conducting more in-depth and detailed tests under various changing network conditions. These tests will include varying the maximum number of child nodes per parent and exploring specific scenarios and edge cases to thoroughly evaluate MTP's performance and robustness. By systematically altering these parameters, we aim to understand the limits and potentials of MTP in different environments and ensure its reliability and efficiency across a wide range of applications. This will help in refining the protocol and uncovering new insights that can further improve network communication strategies. By increasing the maximum child count for parent nodes, group creation time can be decreased; however, this approach has a drawback during data transmission, as parent nodes would need to handle a significantly higher volume of child node data. That is why our protocol has a dynamic child count parameter, allowing users to adjust this count to suit their specific needs and optimize performance for their particular use cases.

6 RESPONSIBILITIES

Atakan Öztarak was responsible for drafting the initial design and architecture of the system and developing additional features. His contributions have been pivotal in defining the technical architecture.

Cem Meriç Şefikoğulları focused on defining the system requirements and specifications and implementing the core functionalities of the system. His work was crucial in setting the project's foundation and ensuring that the system met the specified requirements. Both members contributed equally to the literature review, finalizing the design document with feedback. This collaborative effort

is the foundation of this research and technical implementations throughout the project.

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